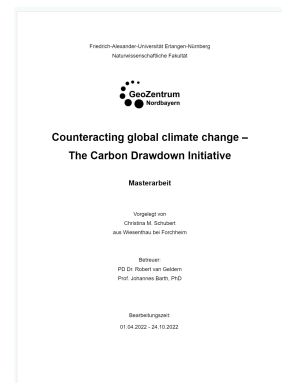




XXL Lysimeter Experiment: Scientific documentation of the first three months (Master Thesis)

May 17, 2023

The building process of our [XXL Lysimeter project](#) in May 2022 and the monitoring of the first 3 months was closely documented by Christina Schubert in her Master Thesis “Counteracting global climate change – The Carbon Drawdown Initiative”.



A PDF file of this master thesis is available at the link below.
We thank Christina Schubert, Dr. Robert van Geldern, Prof. Johannes Barth and the University of Erlangen for allowing us to post this document here.

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In April 2022, the XXL-lysimeter project was started by the Carbon Drawdown Initiative on a field in Fürth to test the CO₂ sequestration capacity of enhanced weathering (EW). Out of twenty self-build lysimeters, four replicates each were applied with no (blank), 100, 200 and 400 t ha⁻¹ coarse basalt, while 200 t ha⁻¹ fine basalt generated three additional lysimeters and a 200 t ha⁻¹ coarse-fine basalt mix a mixed-200 lysimeter. To determine if EW occurs, gas and solute samples were periodically taken during the three-month sampling period. Within the first three weeks after the basalt application an EW signal was confirmed in the gas samples with decreasing soil and tank CO₂ concentrations of up to 32% and increasing stable carbon isotope values ($\delta^{13}\text{C}_{\text{CO}_2}$). These parameters could be potential EW-proxies. Furthermore, the comparison of the CO₂-concentration in the fine and coarse basalt lysimeters confirmed a slightly higher weathering rate with an increased grain surface. Additionally, the weathering signal was looked for in the solute samples via the field parameters, alkalinity, dissolved inorganic carbon (DIC), stable carbon isotopes ($\delta^{13}\text{C}_{\text{DIC}}$), metal and trace metals and the water isotopes ($\delta^2\text{H}$ and $\delta^{18}\text{O}$). The measured general increases in alkalinity, DIC, electrical conductivity (EC), metals and trace metals and the pH of all lysimeters are likely mostly induced by rising amounts of tap water irrigation during the drought period. Otherwise, no distinguished basalt trends, compared to the blank lysimeters, were detected, except for the mixed-200 lysimeter. This could be due to mixing and sorption processes in the soil, which was deduced from the water isotopes and the lysimeter water output in relation to the maximal soil pore space. Only the mixed-200 lysimeter has a significantly higher water output than the maximal pore space, so that younger input water from after the basalt application, with a possible EW signal, is measurable. This could explain the higher values, compared to the blank, for alkalinity and DIC from second to fifth sampling with a decreasing $\delta^{13}\text{C}_{\text{DIC}}$ with higher basalt amounts, an increasing and after sampling two decreasing pH and generally elevated Fe, Cr, Zn, Co, and major cations. Nonetheless, without further samplings no clear confirmation of a possible weathering signal in the mixed-200 lysimeter can be given. Also, a comparison of the mixed-200 lysimeter with a column-lab experiment in Hamburg showed similar trends in DIC and $\delta^{13}\text{C}_{\text{DIC}}$. Lastly, a slight fertilizing effect in the 100 t ha⁻¹ basalt lysimeters was confirmed, that became evident through a higher blossom amount than in the blank lysimeters.

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